

Group 13

PMICs Final Project

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COLUMBIA ENGINEERING

The Fu Foundation School
of Engineering and Applied Science



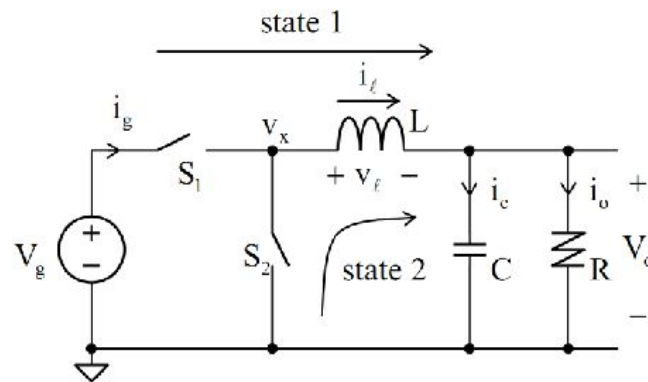
Introduction



Specifications

Buck Converter - Efficient on-chip architecture to stepping 3.3V IO to 1.0 V digital rails, while managing high current load.

- Maximum Current Load: 3A
- Switching Frequency: 1MHz
- Inductor: $1\mu\text{H}$, $15\text{ m}\Omega$
- Capacitor: $10\text{ }\mu\text{F}$, $5\text{ m}\Omega$
- Efficiency Goal: $\geq 90\%$ across load range 2-3A
- Utilizing 3.3V 180nm Library



Architecture

Blocks & Responsibilities

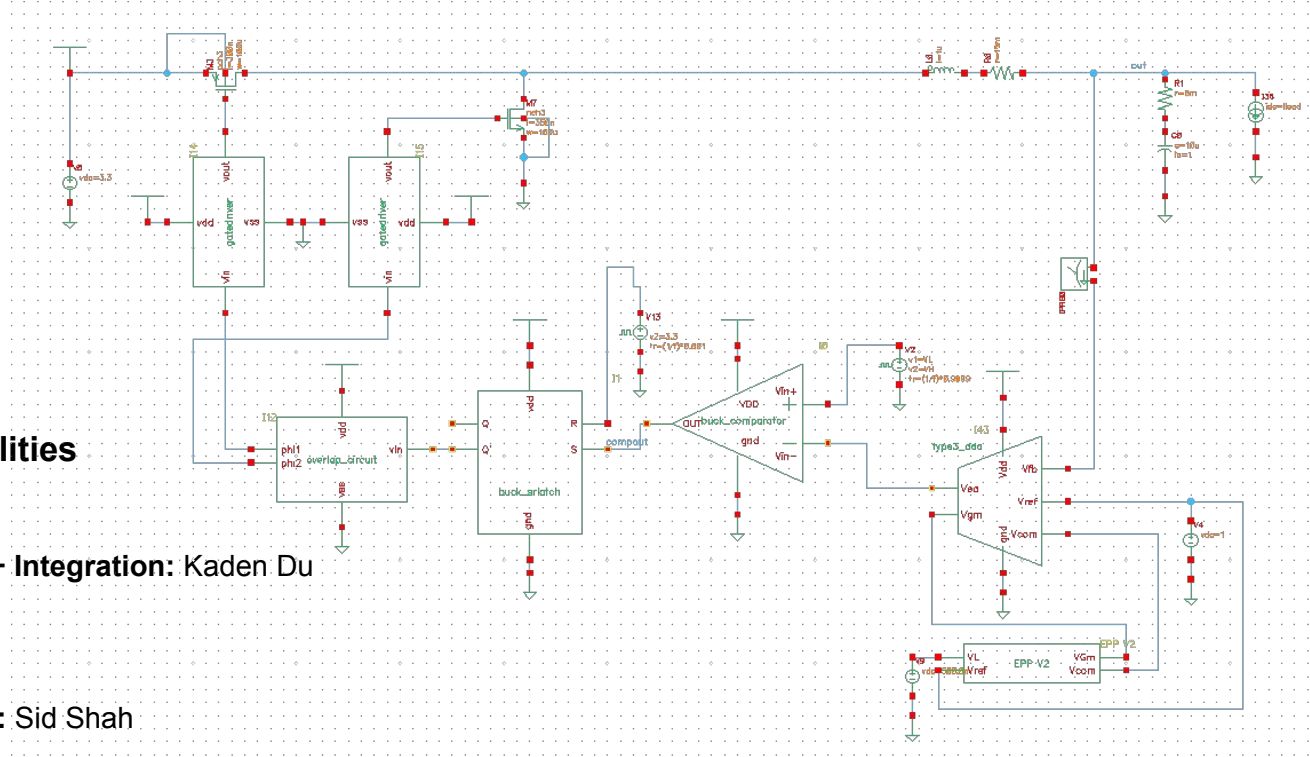
DDA: James Jagielski

Comparator + SR Latch + Integration: Kaden Du

OTA: Logan Po

EPP: Gabriel Tetrault

Gate Driver + DT Control: Sid Shah



Individual Block Design

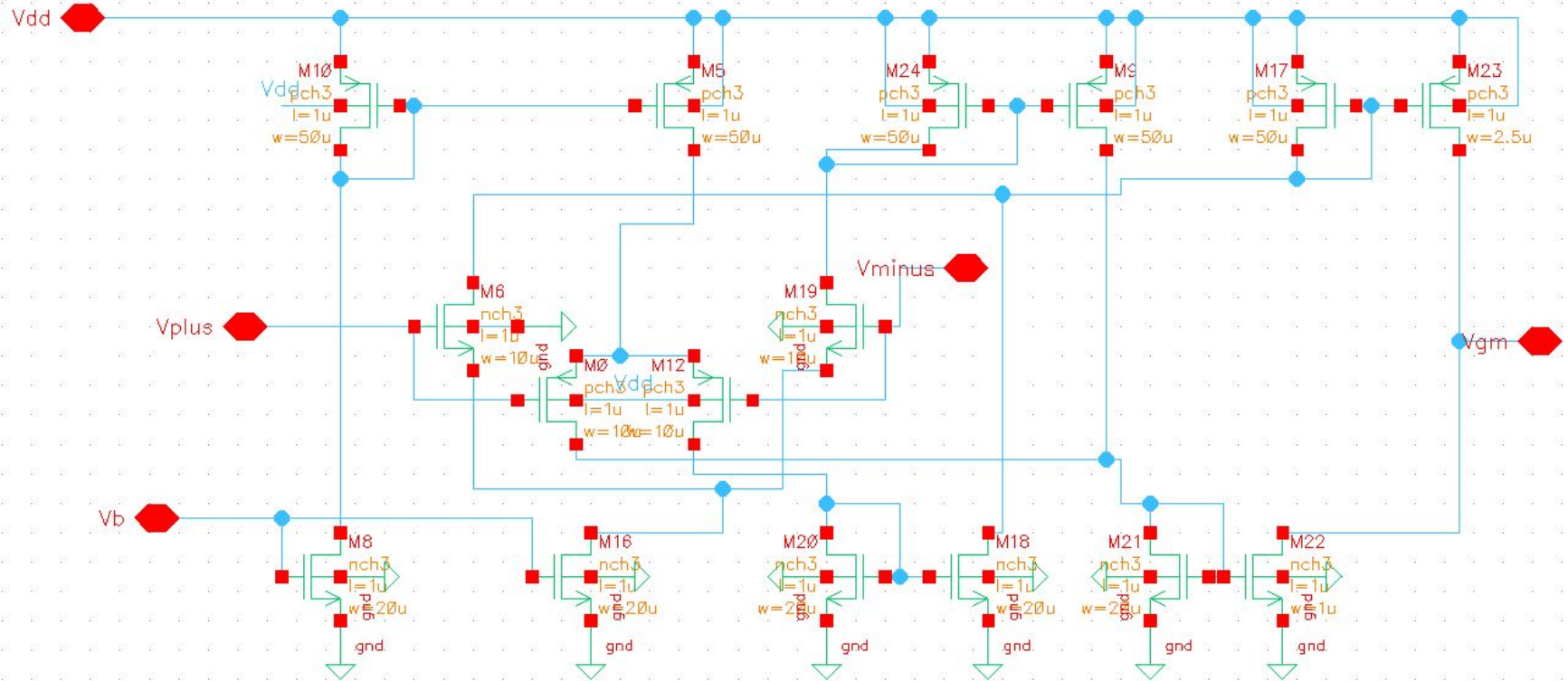


LC sizing

- Used 1uH inductor to mitigate ripple
 - Modern parts such as Vishay Dale IHL2000VZEB1R2M31 report a series resistance of 15mOhm
- 0402 Capacitors report 5mOhm Series Resistance

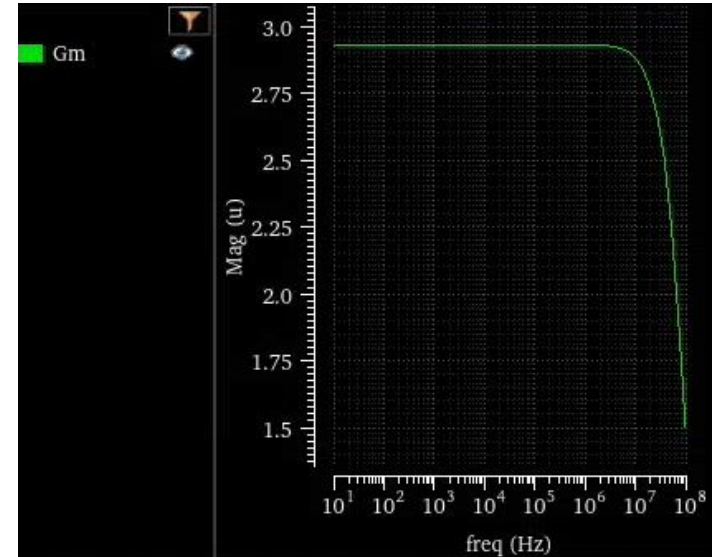
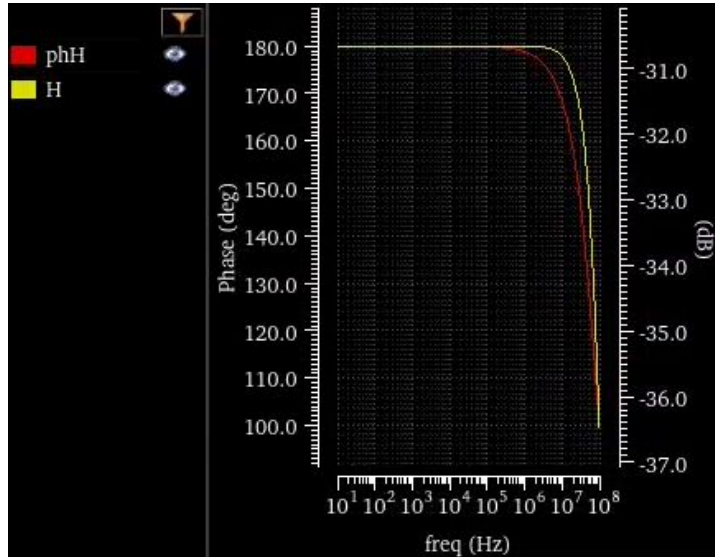


Operational Transconductance Amplifier (OTA)



OTA

Performance Metrics

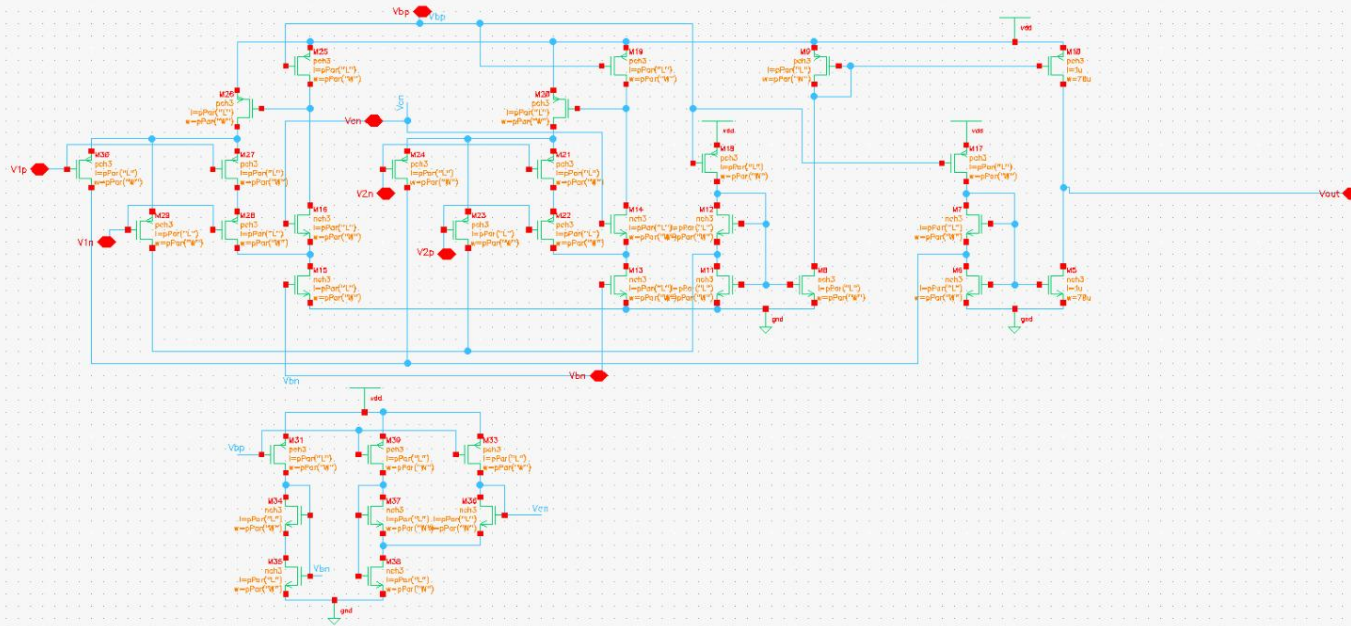


Gm gain: 2.93 μ S

Bandwidth: 28 MHz for 1 dB; 61 MHz for -3 dB

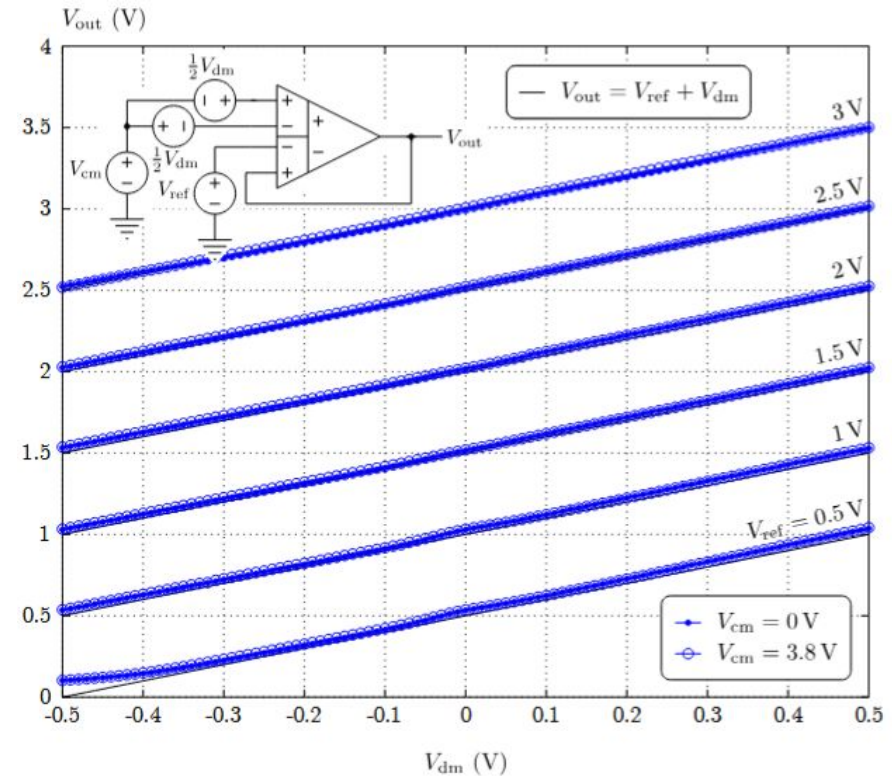
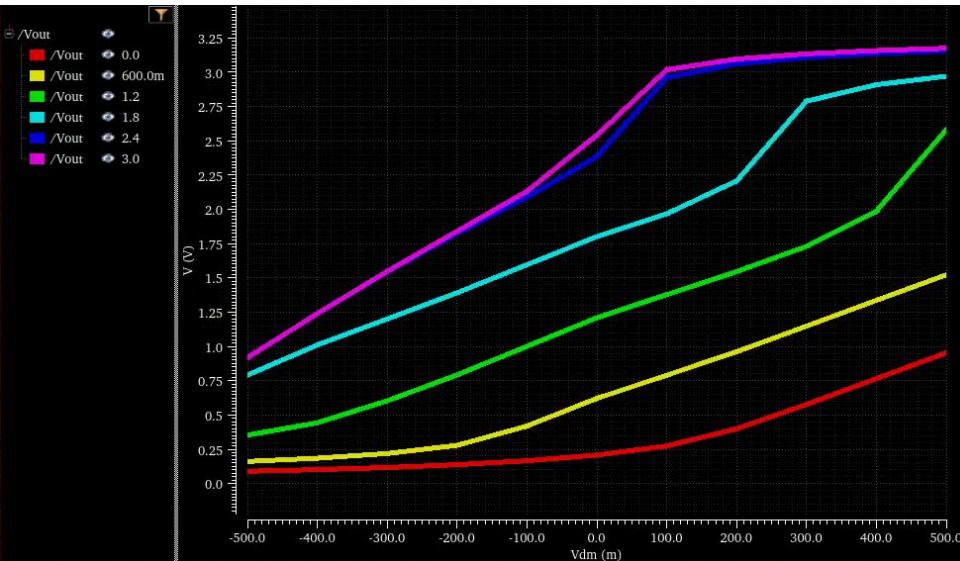
(Output resistance $R_o = 10$ k Ω)

Differential Difference Amplifier (DDA)

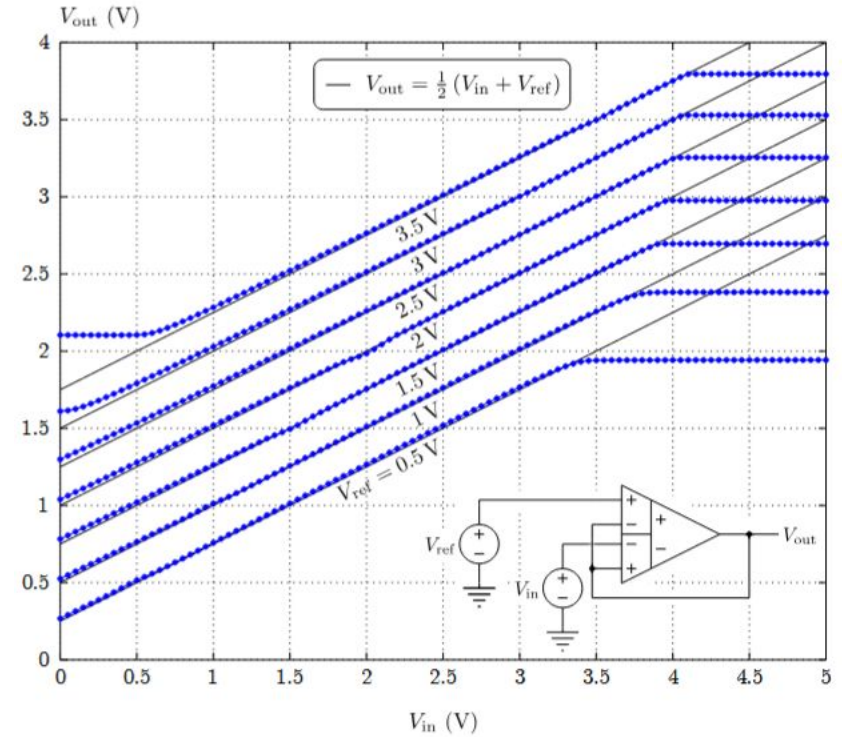
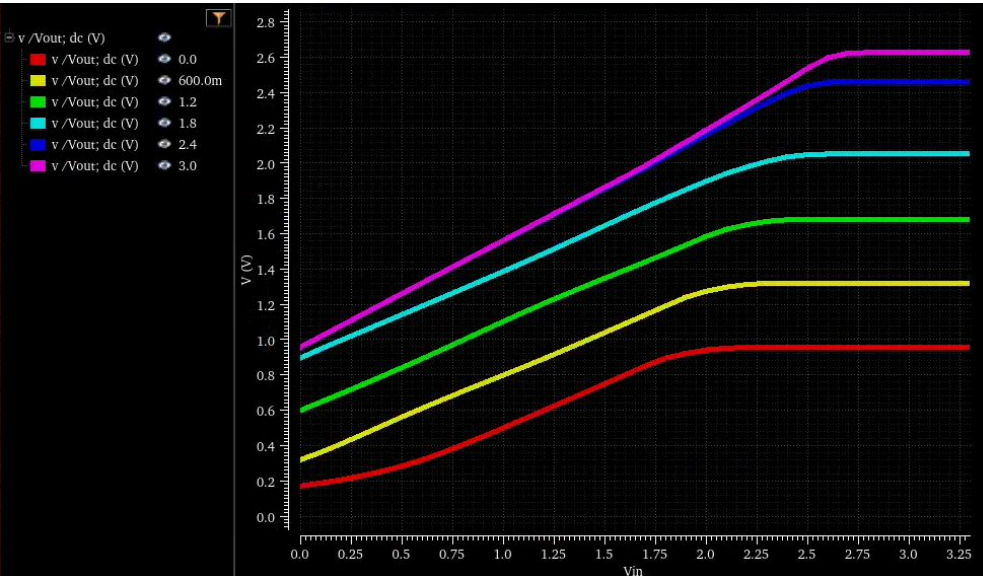


CMOS differential-difference amplifier (DDA) circuit with a pair of matched class-AB input stages that each produce a differential output current.

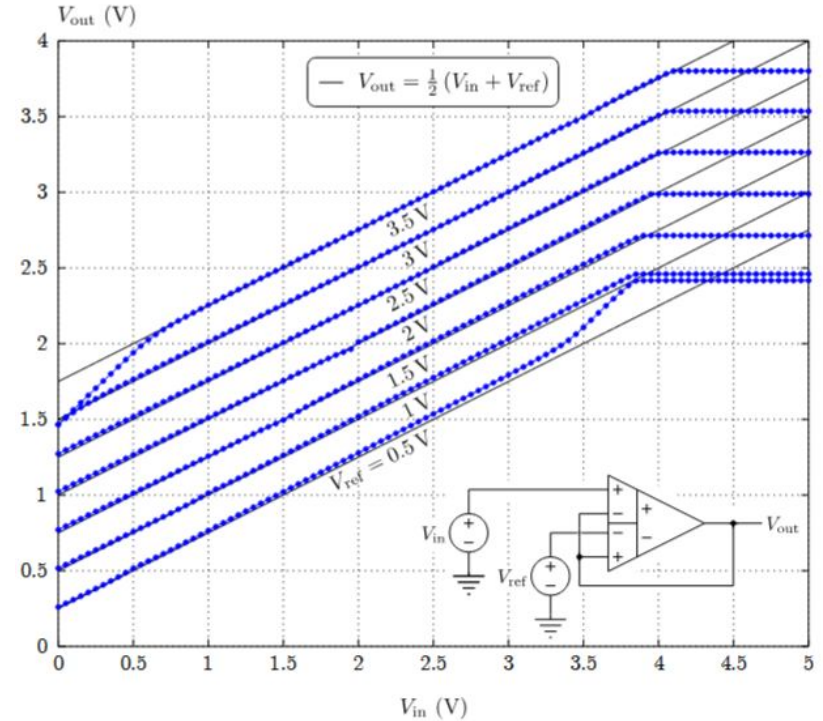
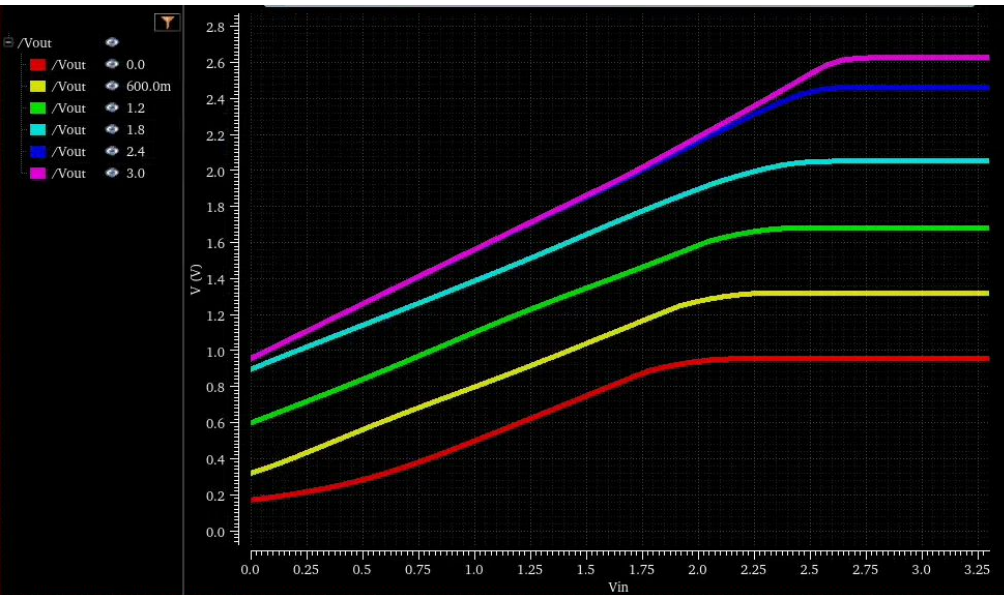
Balun



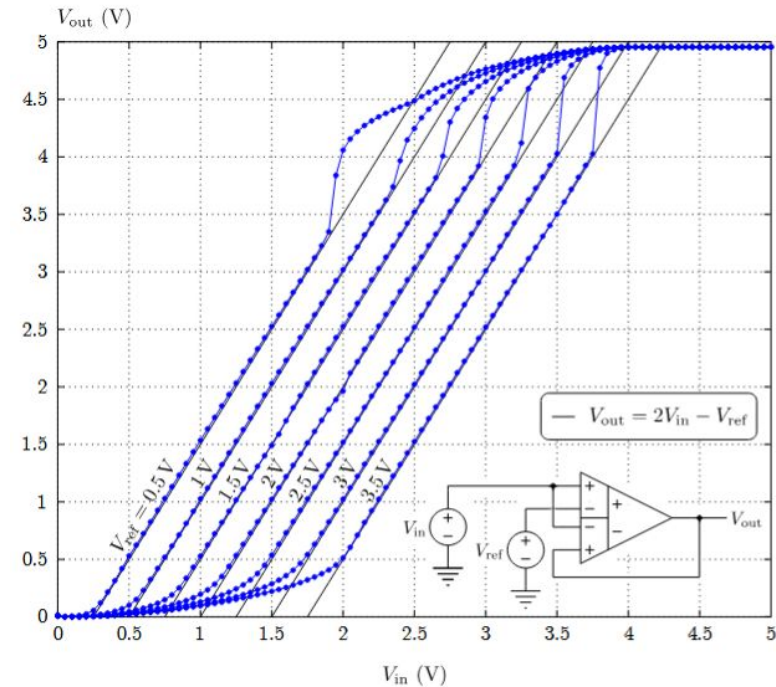
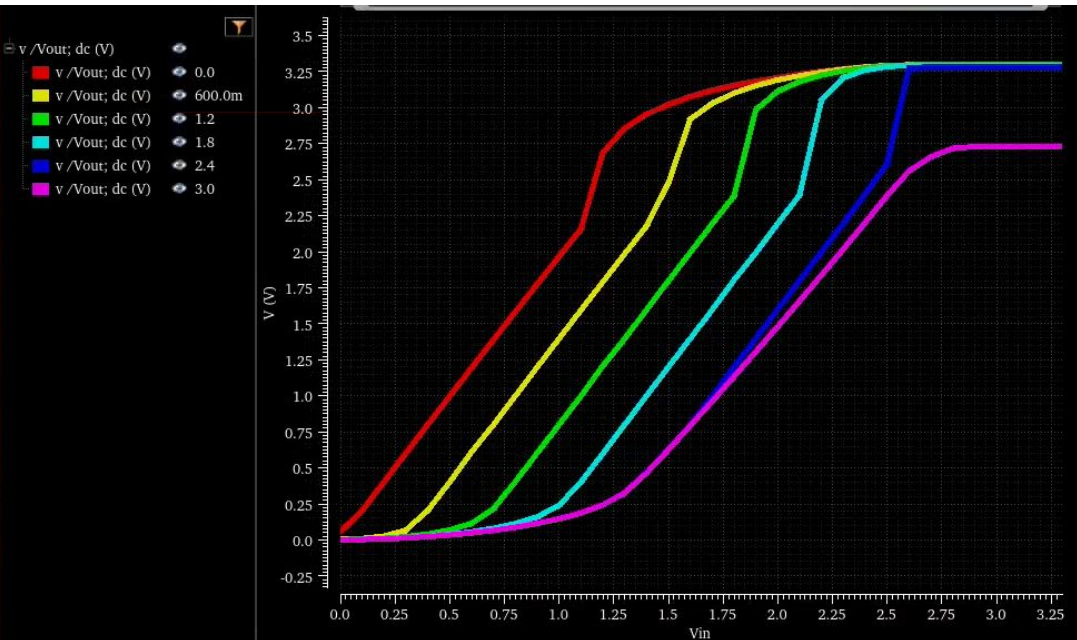
Common Mode detector



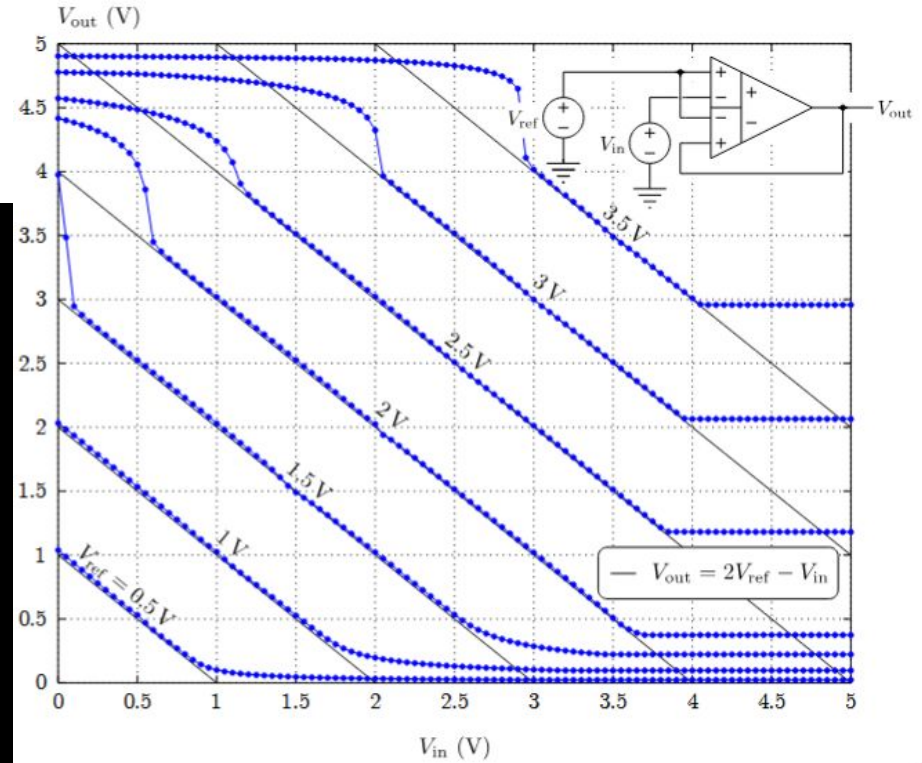
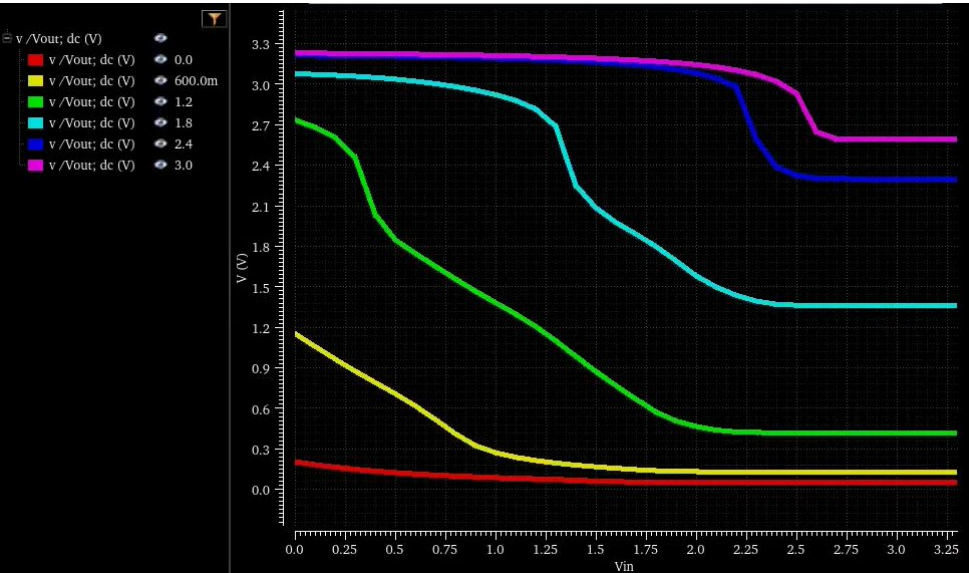
Averager



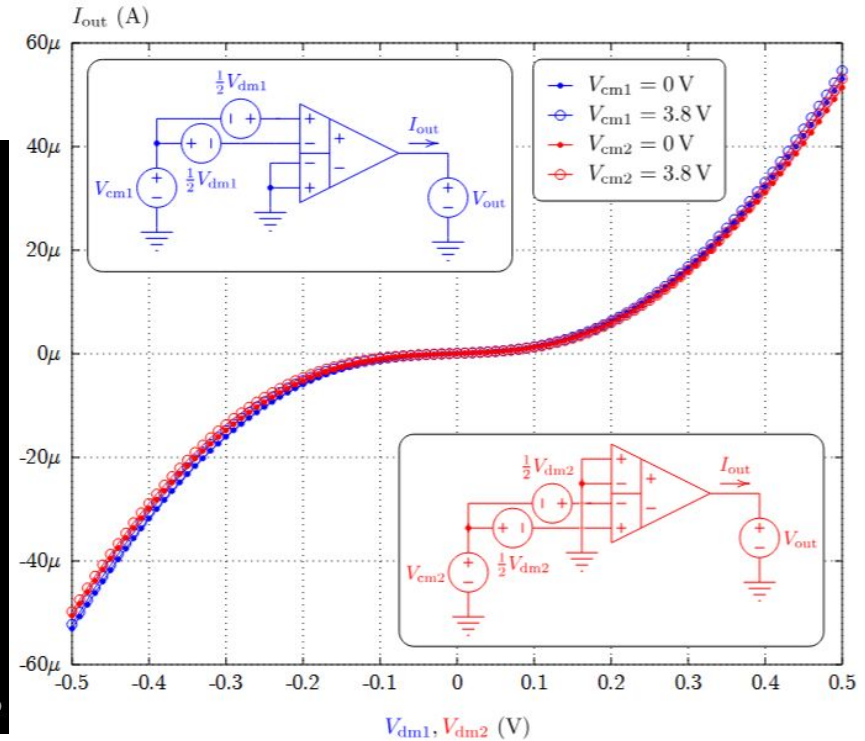
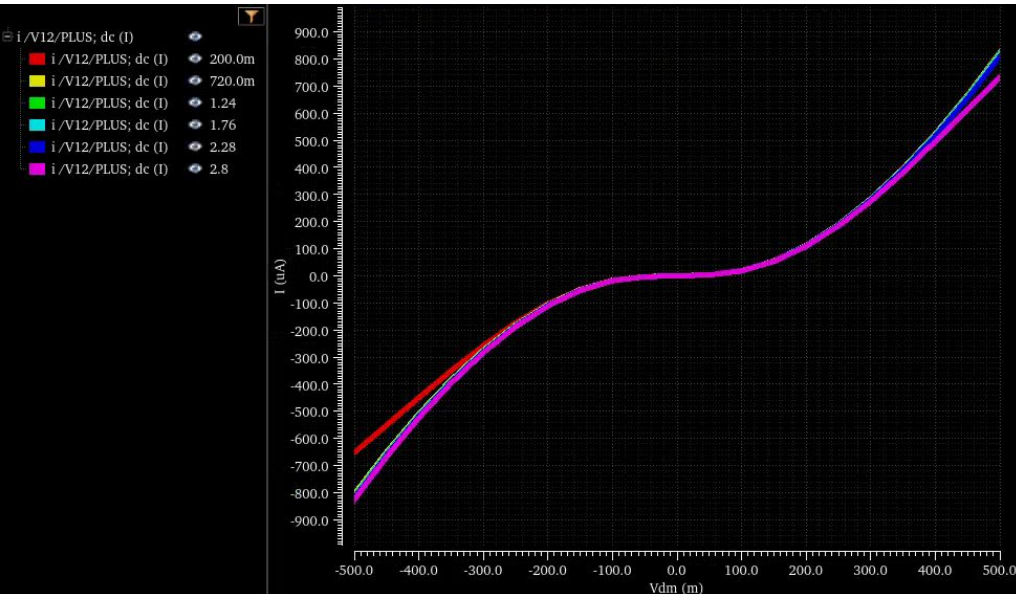
Doubler



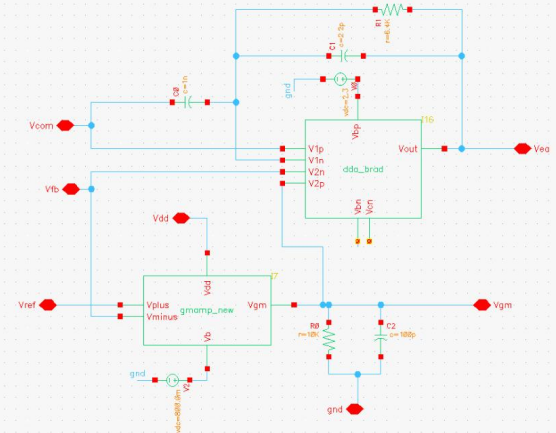
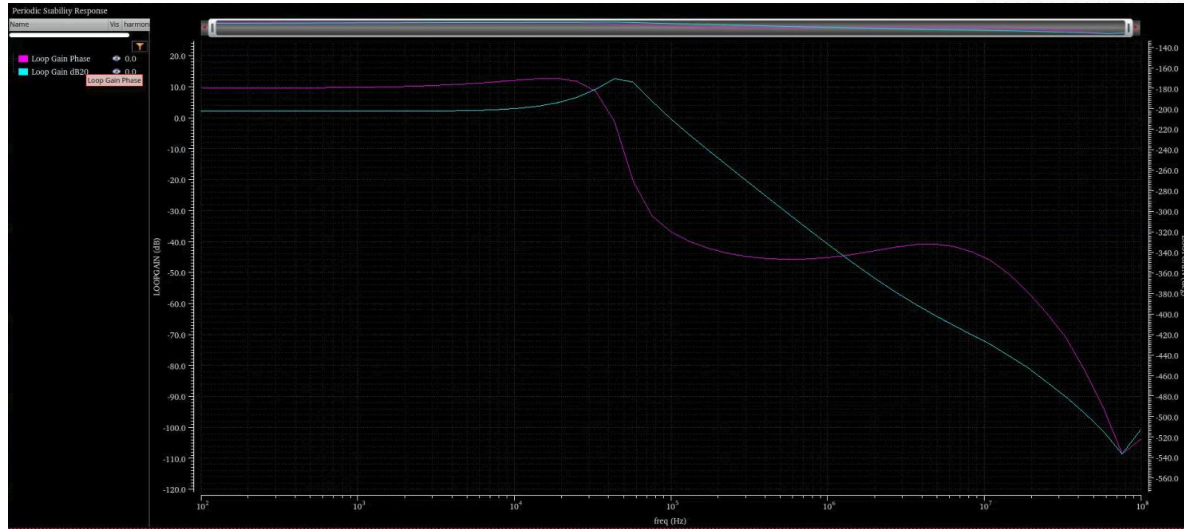
Inverter



IV characteristics



Type III Compensation

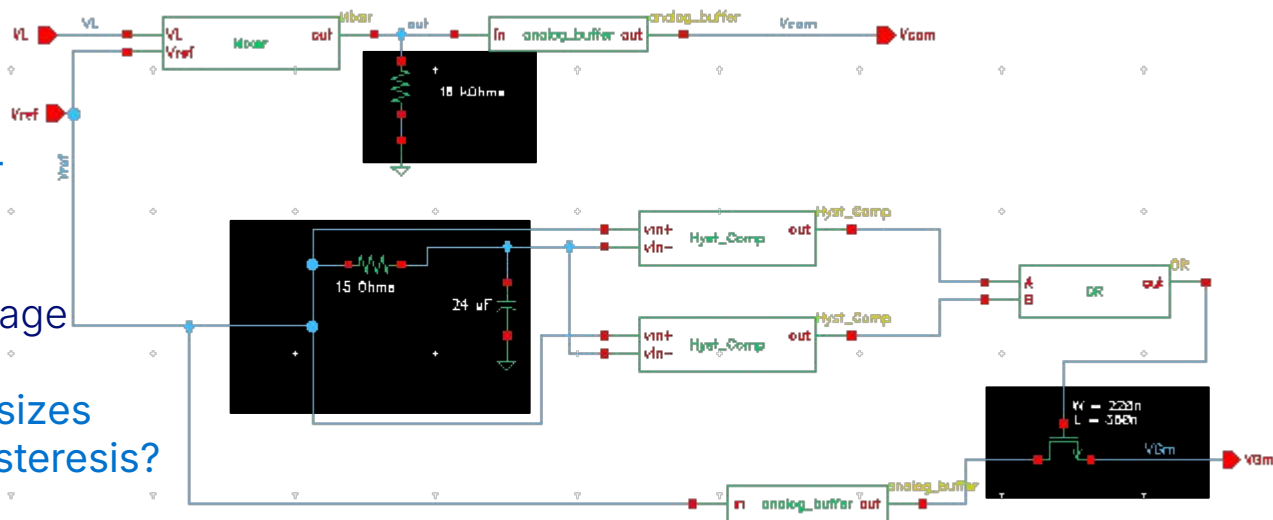


End-Point Prediction (EPP)

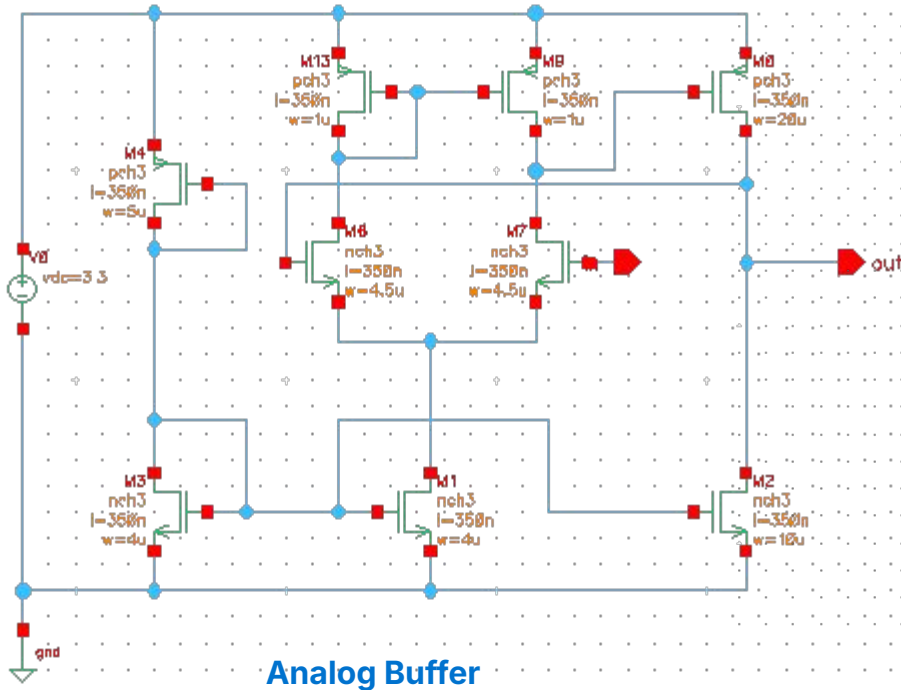
- Two sub-circuits
 - Vcom current mixing (top)
 - Vref-VGm tracking (bottom)

$$V_{com}(V_{1+}) = V_{ea} = \frac{V_m}{bV_g} V_{ref} + V_L$$

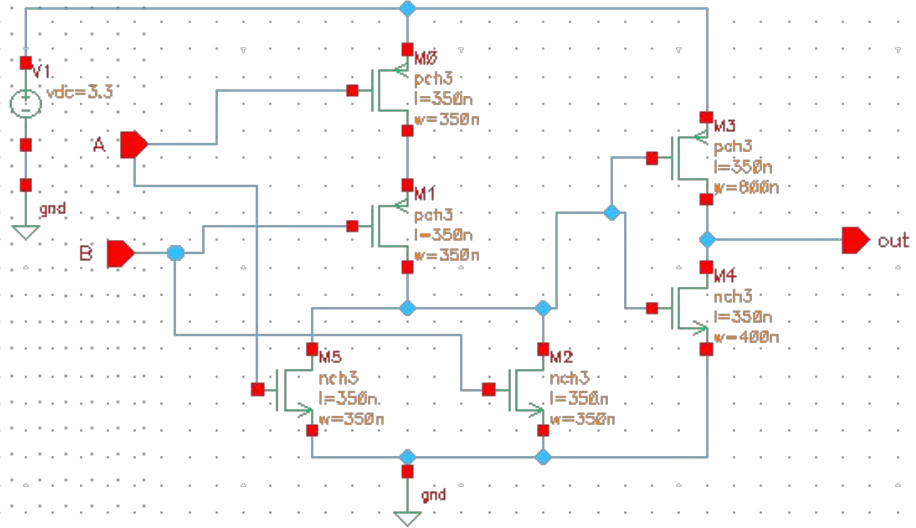
- Trade-Off 1: Mixing resistor
- RC LPF produces delay
- Two comparators check
 - One detects undervoltage
 - Other for overvoltage
- Trade-Off 2: RC delay and sizes
- Trade-Off 3: How much hysteresis?
- Trade-Off 4: Switch sizing
- Buffer and OR gate built as expected



End-Point Prediction (EPP) - Analog Buffer and OR Gate



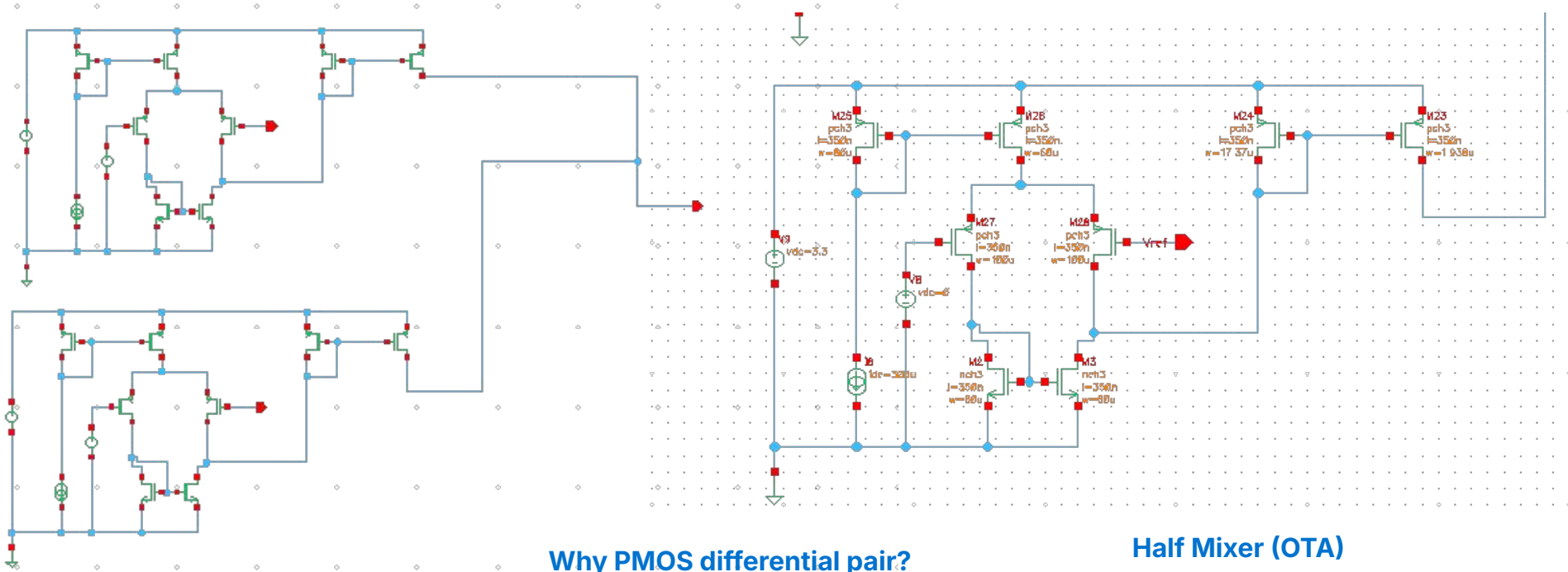
Analog Buffer



OR Gate



End-Point Prediction (EPP) - Current Mixer



Overall Mixer

Why PMOS differential pair?

Half Mixer (OTA)

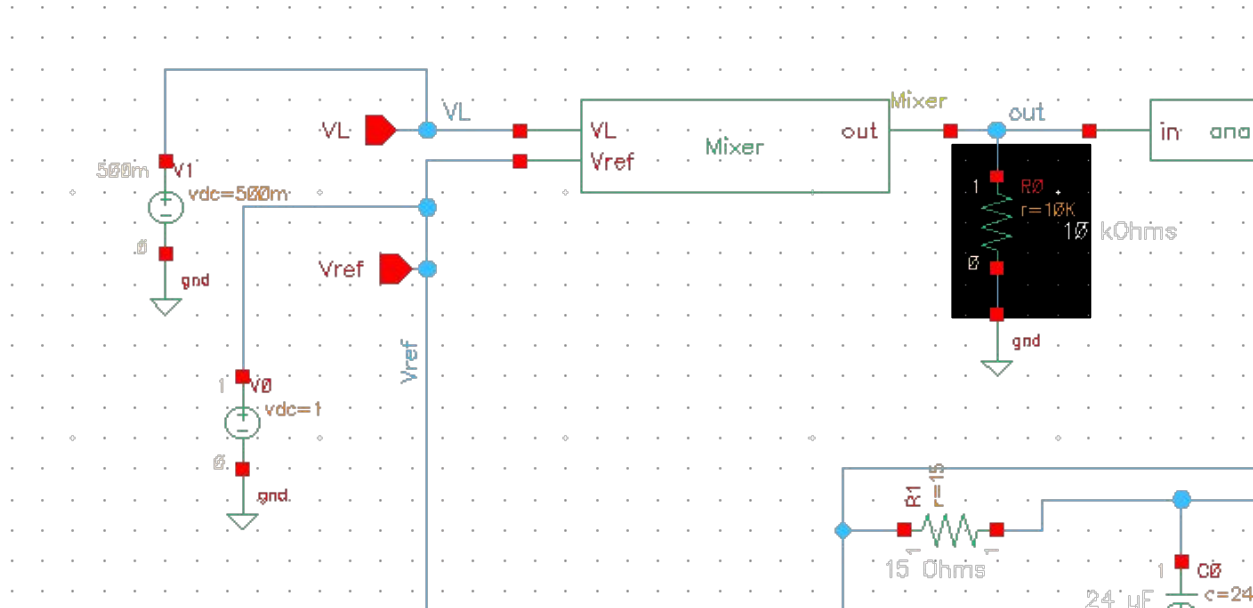
Why current mirror output?



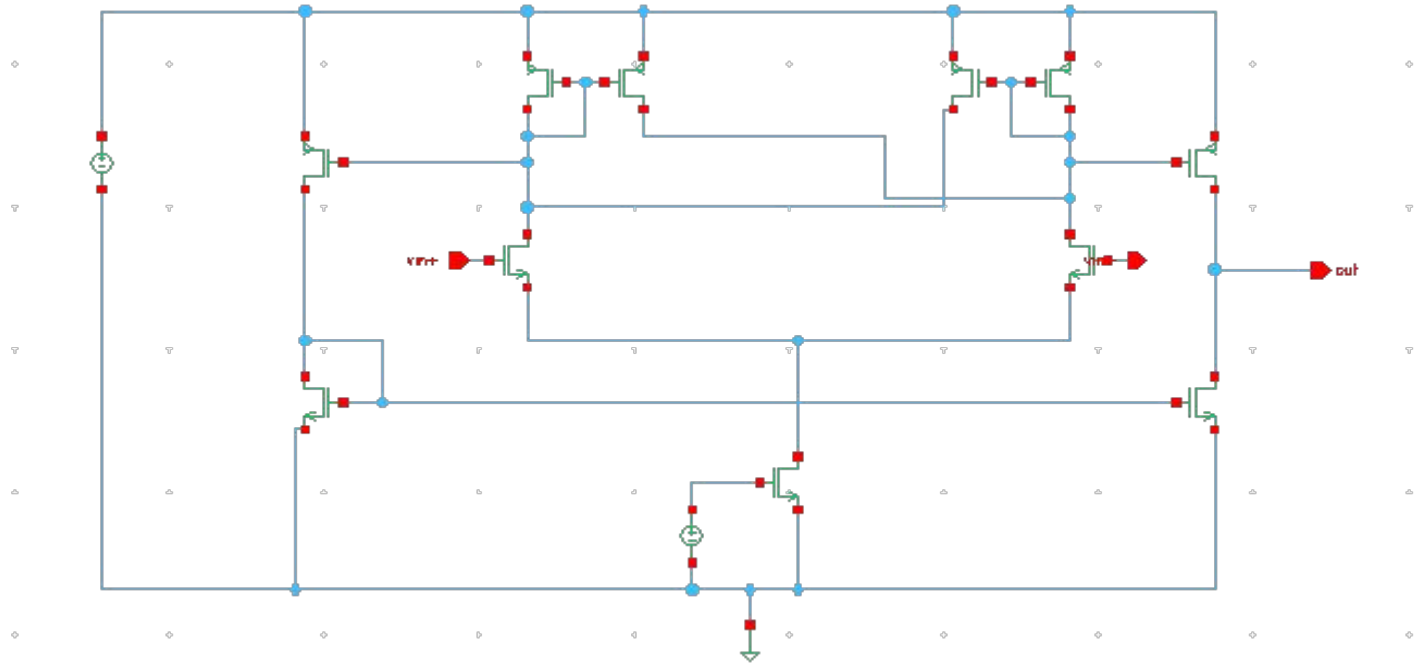
End-Point Prediction (EPP) - Current Mixer Results

Example: $V_H = 2.15\text{ V}$, $V_L = 0.5\text{ V}$

$$V_{\text{com}}(V_{1+}) = V_{ea} = \frac{V_m}{bV_g} V_{\text{ref}} + V_L$$

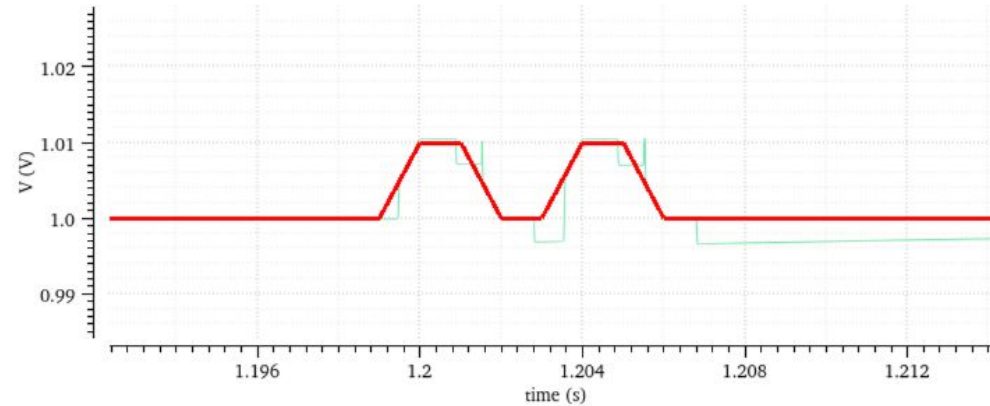


End-Point Prediction (EPP) - Hysteretic Comparator

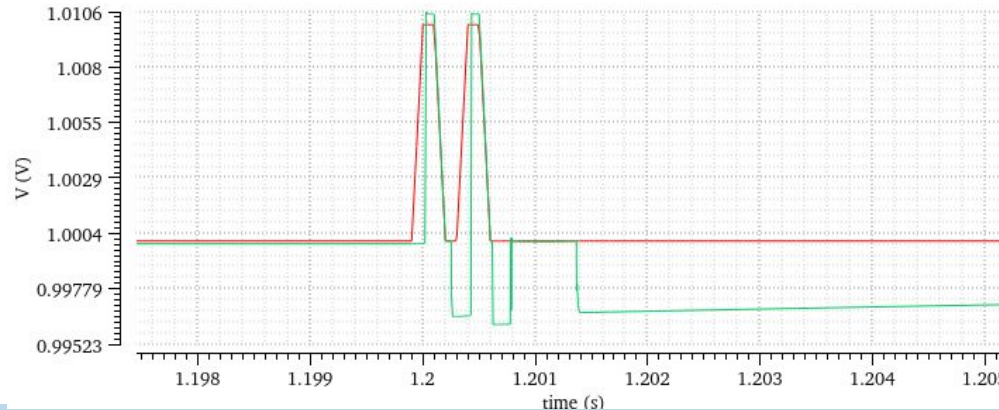


End-Point Prediction (EPP) - Hysteretic Comparator Results

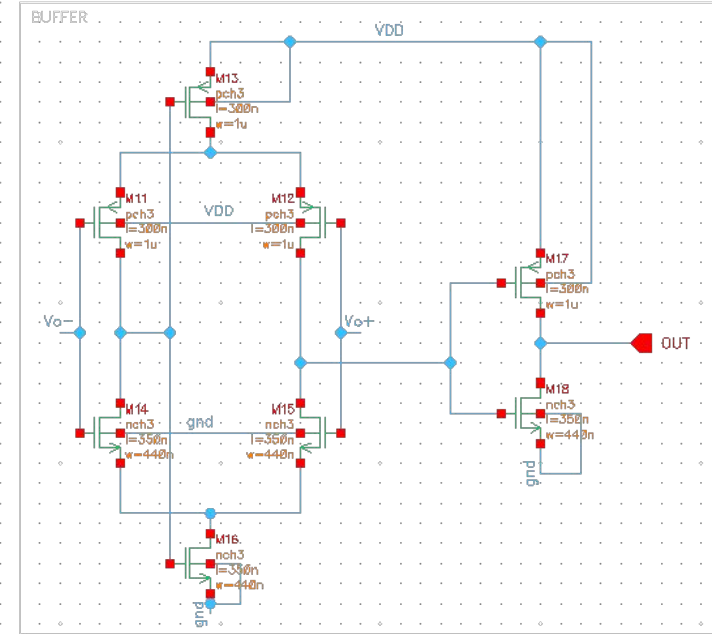
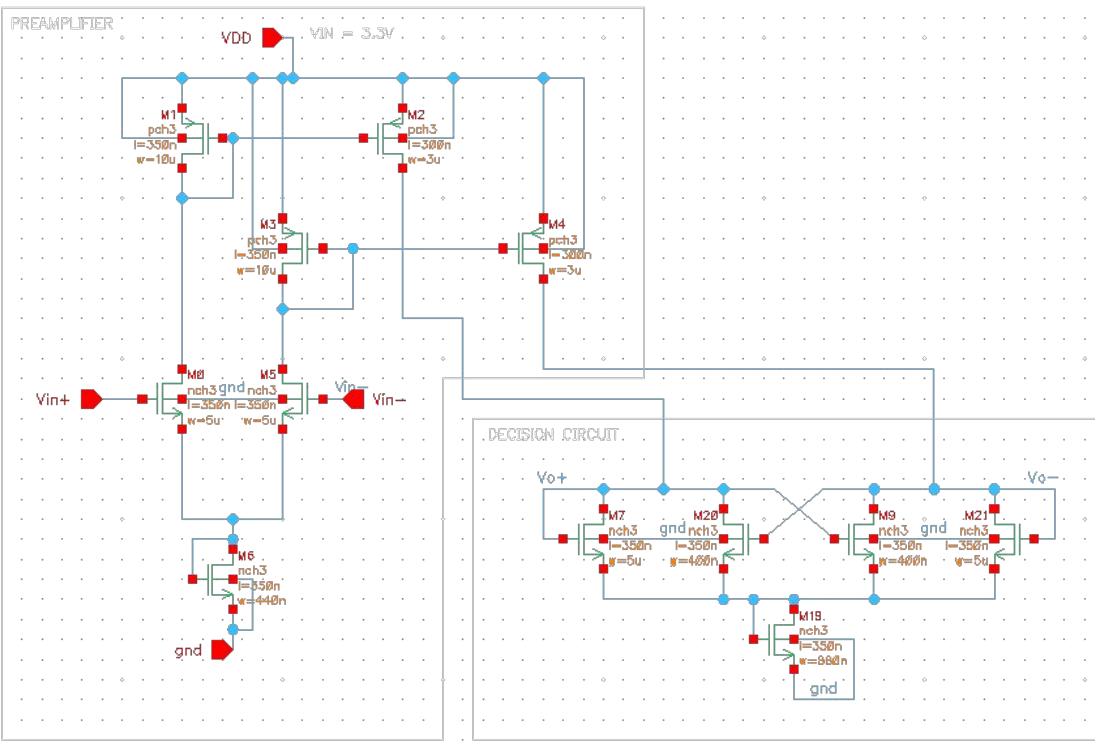
1 ms scale fluctuations
Tracks within 10 mV



100 us scale fluctuations
Tracks within 10 mV



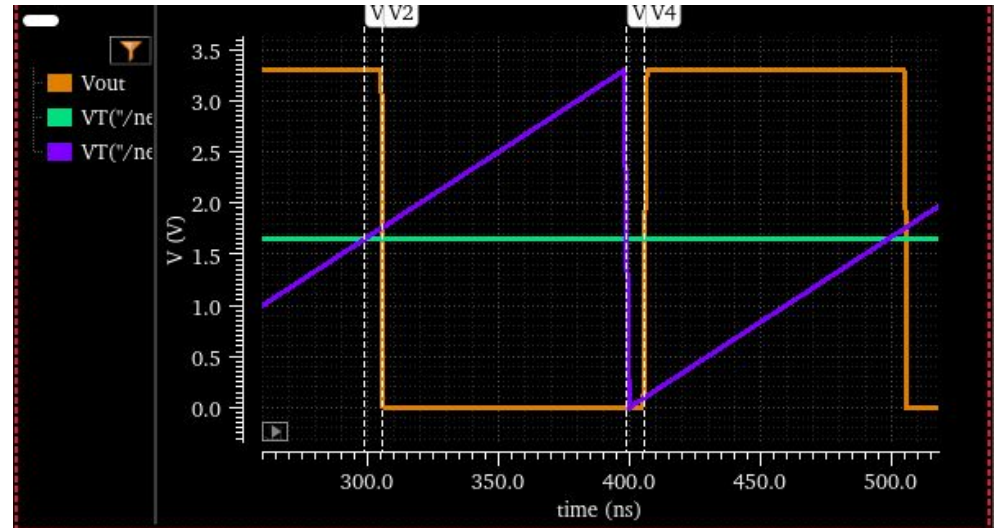
Comparator



Comparator - Results

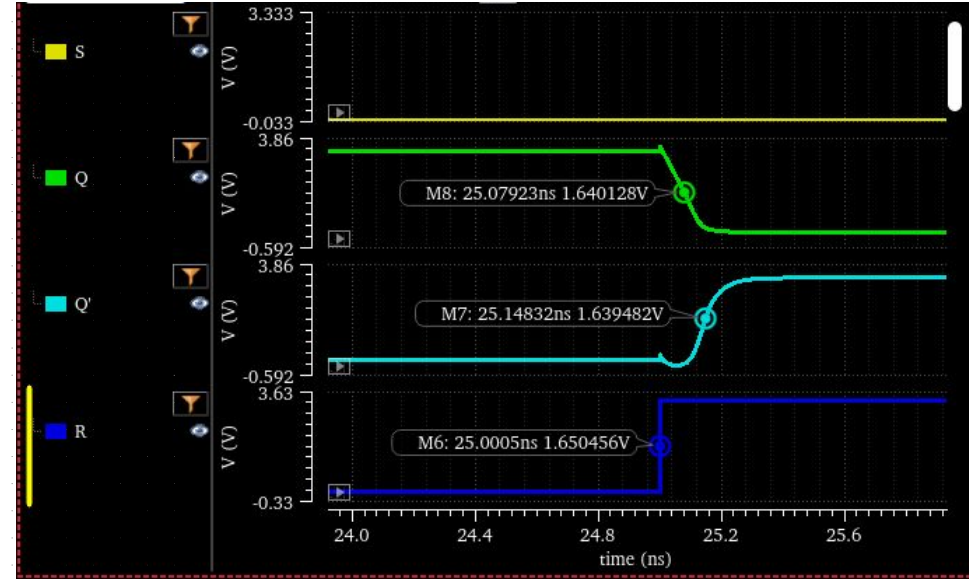
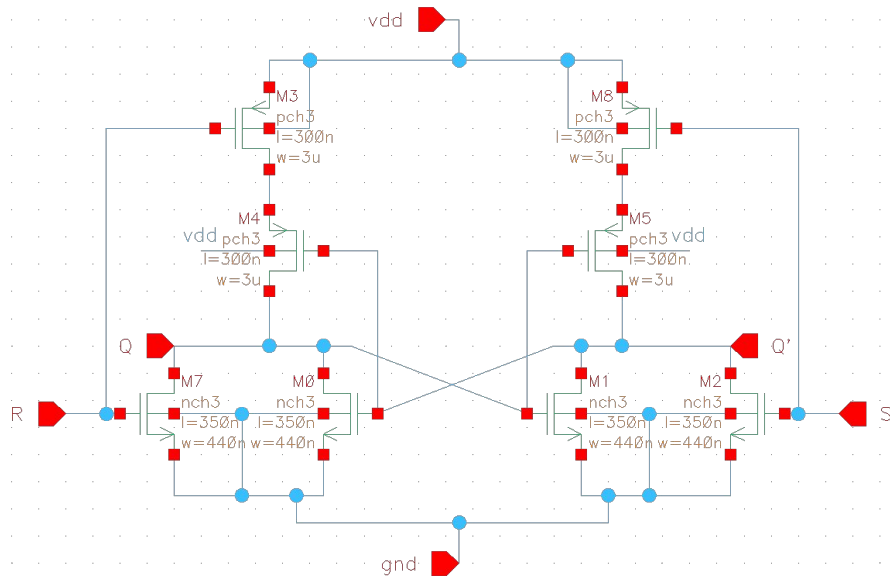
Optimized for:

- Delay - $\sim 5\text{ns}$
- Power Consumption: $\sim 238\mu\text{W}$



SR Latch

- NOR Based Latch
- Optimized response time + low power consumption
- Delay = 0.15ns



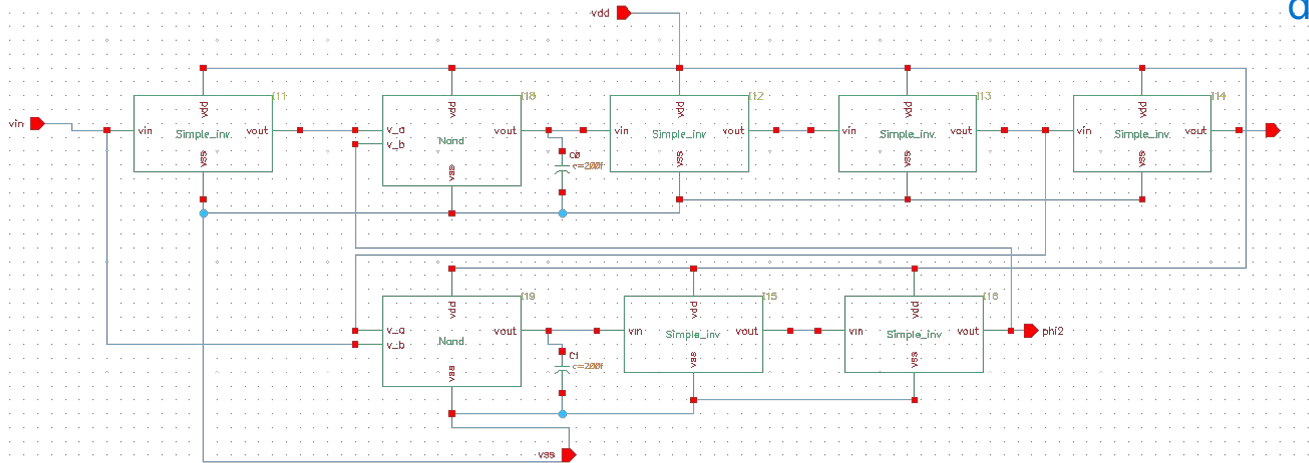
Delay: ~0.15ns

Deadtime Control

In a synchronous buck, having both PFETs turn on simultaneously → shoot through loss.

Non-overlapping circuit uses a large capacitor + inverters to control delay.

Using passive components (cap) reduces inverter driver consumption while setting more precise deadtime



$$t_p = \frac{C_L V_{DD}}{I_{drive}}$$



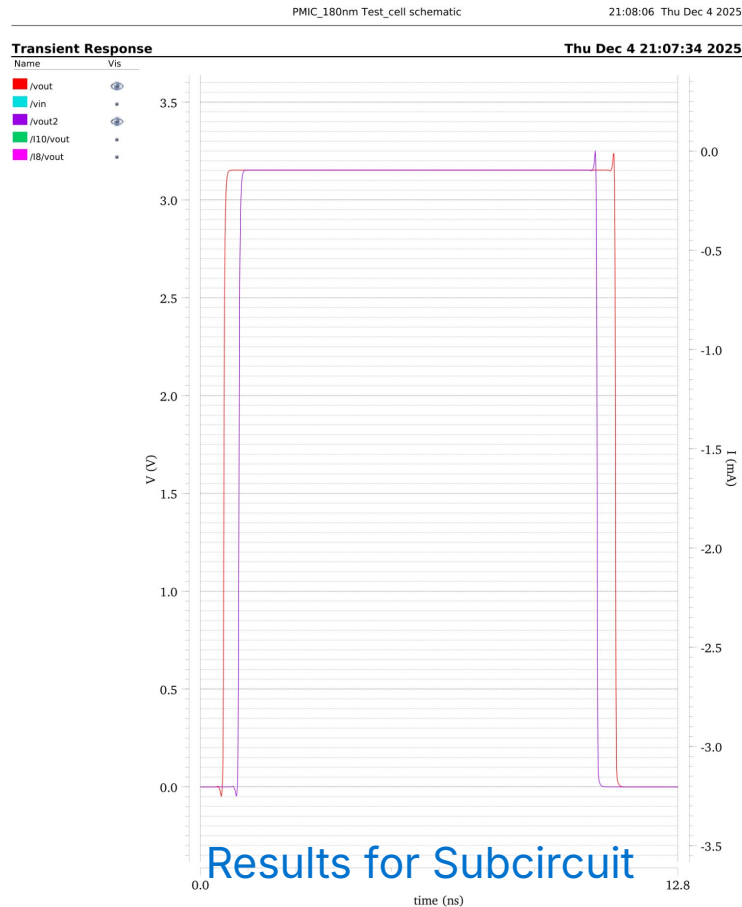
Deadtime Control - results

Dead-time control was refined by performing a capacitor sweep

1.75pF was found to be best value for capacitor, giving the converter a 2.5n DT.

In a 500ns period, this gave 0.5% optimal deadtime

Overall delay of circuit was <3ns, well under transient specifications.



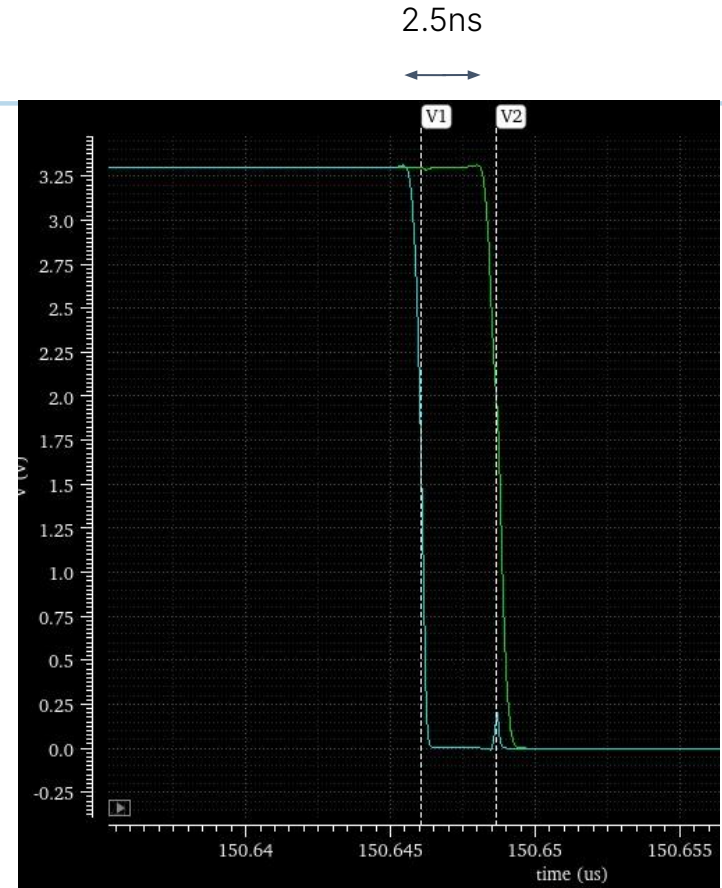
Deadtime Control - results

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PFET Gate Driver

Driving the PFETs - We need enough capacitance

We want to minimize switching loss and reduce driving loss

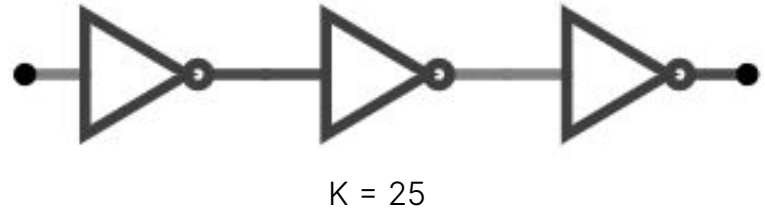
Since efficiency is more greatly valued, a higher k was chosen (25)

Resulted with 49mW & 34mW power draw for the PMOS and NMOS driver with <10ns delay

Due to electron/hole mobility, the PMOS sizing was 2x NMOS.

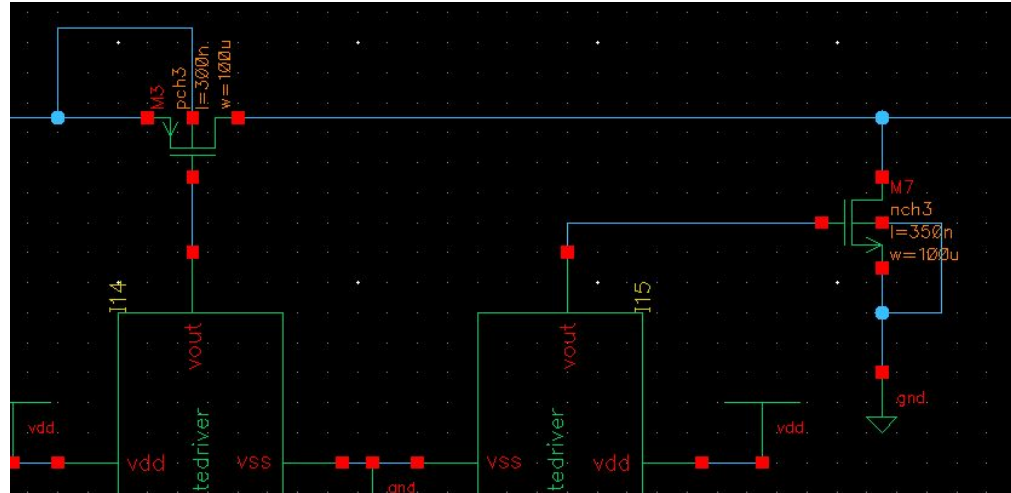
$$P_{sw} = \frac{1}{2} V_{DS} I_D (t_r + t_f) f_{sw}$$

$$P_{gate} = (1 + \frac{1}{k}) C V^2 f_{sw}$$



Transistor Sizing

- Transistors were sized to compromise loss with necessary driver size
- PMOS is sized at 100u W, 300n L, and M=20000 for a total area of **0.006 cm²**
- Nmos is half this size, **0.003 cm²**
- PMOS loss is 36mW at 3A load
- NMOS loss is 14mW at 3A load

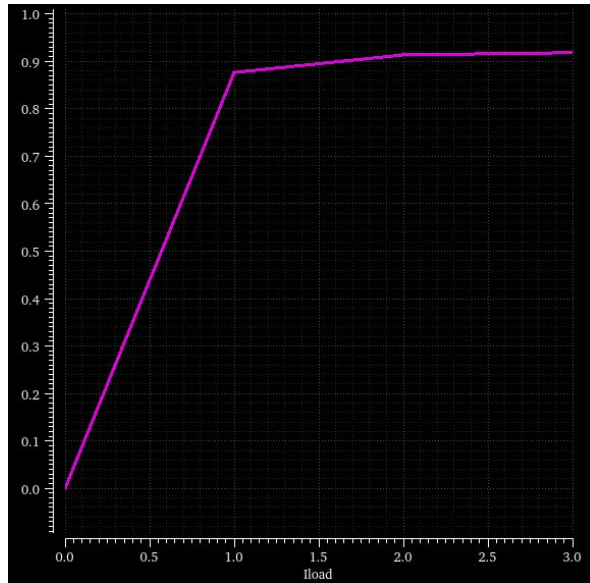


Results

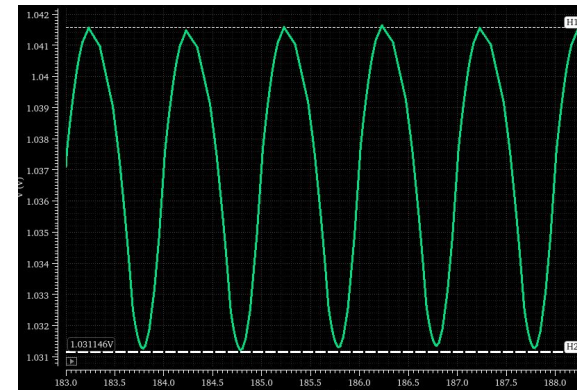
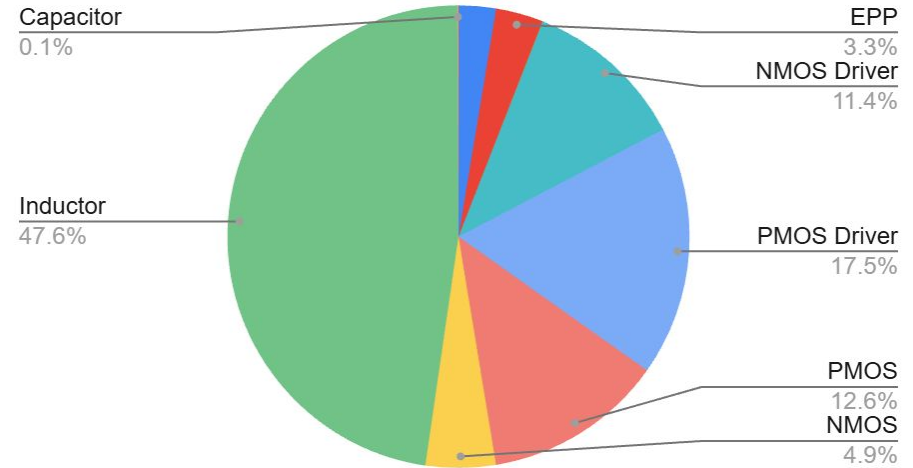
Steady State Response

- Regulates to 1.035 V
- 10mV steady state ripple (3A load)

Efficiency across load:



Power Losses at 3A load



Transient Response - Load and Line Step

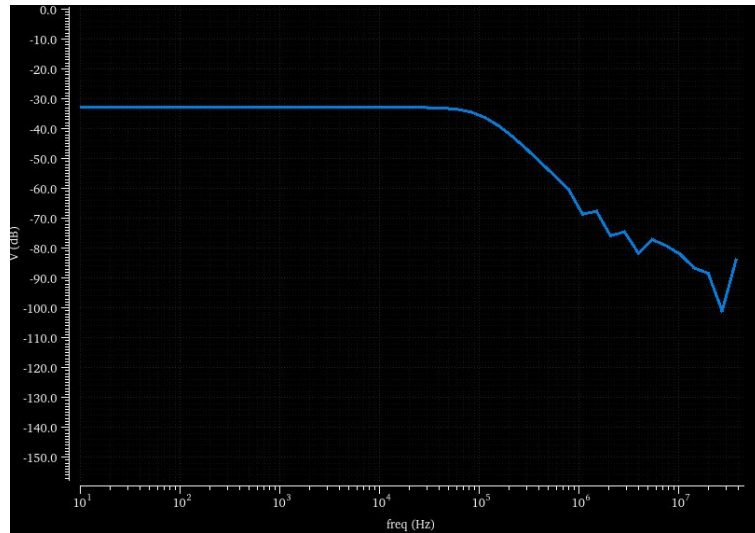
- 70mV undershoot, 130mV overshoot on 2A-3A step



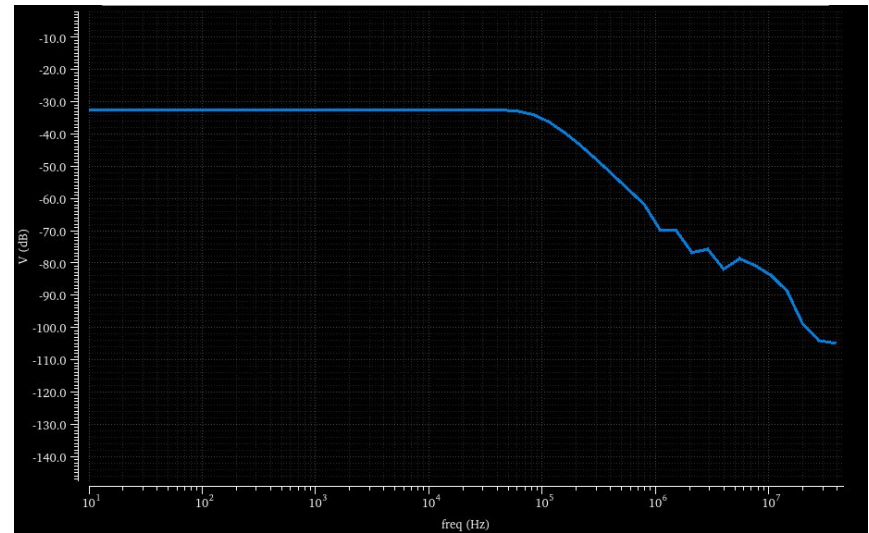
Power Supply Rejection Ratio

- PSRR tested through PAC simulations

PSRR @ 0.1A Load (-32.8dB at DC)



PSRR @ 3A Load (-32.6dB at DC)



Thank you!



References

[1] L. Cheng, Y. Liu and W. -H. Ki, "A 10/30 MHz Fast Reference-Tracking Buck Converter With DDA-Based Type-III Compensator," in IEEE Journal of Solid-State Circuits, vol. 49, no. 12, pp. 2788-2799, Dec. 2014, doi: 10.1109/JSSC.2014.2346770.

